

# The Spectrum of an Operator

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# Outline

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Examples of Spectra of Operators

## Definition

*Let  $T$  be a linear operator,  $\lambda \in \mathbb{C}$  is called an eigenvalue of  $T$ , if there exists  $x \in H$  with  $x \neq 0$  such that*

$$Tx = \lambda x$$

## Theorem

*Let  $H$  be a Hilbert space and  $T : H \rightarrow H$  then  $T$  is invertible  $\iff$   $\ker T^* = \{0\}$  and  $\exists \alpha > 0$  such that  $\|T(x)\| \geq \alpha \|x\| \quad \forall x \in H$ .*

## Theorem

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## Corollary

*Let  $H$  be a Hilbert space and  $T : H \rightarrow H$  then  $T$  is invertible  $\iff \exists \beta > 0$  such that  $\|T^*(x)\| \geq \beta \|x\|$  and  $\exists \alpha > 0$  such that  $\|T(x)\| \geq \alpha \|x\| \quad \forall x \in H$ .*



### Proof.

If  $T$  is invertible, then so is  $T^*$ . So, by Theorem 1,  $\exists \beta > 0$  such that  $\|T^*(x)\| \geq \beta \|x\|$  and  $\exists \alpha > 0$  such that  $\|T(x)\| \geq \alpha \|x\|$  for all  $x \in H$ . Conversely, if  $\exists \beta > 0$  such that  $\|T^*(x)\| \geq \beta \|x\|$ , then if  $T^*u = 0$  for some  $u \in H$ , then

$$\begin{aligned} 0 &= \|T^*u\| \geq \beta \|u\| \\ &\implies u = 0 \end{aligned}$$

So,  $\ker T^* = \{0\}$ . By Theorem 1,  $T$  is invertible. □

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## Definition

*Let  $T$  be a linear operator. The resolvent set of  $T$ , denoted by  $\rho(T)$ , is  $\rho(T) = \{\lambda \in \mathbb{C} : T - \lambda I \text{ is invertible}\}$ . The spectrum of  $T$ , denoted by  $\sigma(T)$  is  $\sigma(T) = \mathbb{C} \setminus \rho(T)$ , or equivalently,  $\sigma(T) = \{\lambda \in \mathbb{C} : T - \lambda I \text{ is not invertible}\}$*

## Remark

*Notice that, if  $T \in B(H)$ , by Theorem 1, we can rewrite the resolvent set of  $T$  as*

$$\rho(T) = \{\lambda \in \mathbb{C} : \ker(T - \lambda I)^* = \{0\} \\ \text{and } \|(T - \lambda I)x\| \geq \alpha \|x\| \text{ for some } \alpha > 0\}$$

*Alternatively, we can rewrite the resolvent set of  $T$  as*

$$\rho(T) = \{\lambda \in \mathbb{C} : \|(T - \lambda I)^*x\| \geq \beta \|x\| \\ \text{and } \|(T - \lambda I)x\| \geq \alpha \|x\| \text{ for some } \beta, \alpha > 0\}$$

*by Corollary 1*

## Theorem

*If  $\lambda$  is an eigenvalue of  $T$  , then  $\lambda \in \sigma(T)$*

### Proof.

Suppose that  $\lambda \notin \sigma(T)$ , then  $T - \lambda I$  is invertible. So, we have  $(T - \lambda I)u = 0 \implies u = (T - \lambda I)^{-1}(T - \lambda I)u = (T - \lambda I)^{-1}(0) = 0$ . So, if there is  $u \in \mathcal{H}$  such that  $Tu = \lambda u$ , then  $u = 0$ . So,  $\lambda$  is not an eigenvalue of  $T$ . □

## Theorem

*Let  $H$  be a finite dimensional Hilbert space, and let  $T : H \rightarrow H$  be linear. If  $\lambda \in \sigma(T)$ , then  $\lambda$  is an eigenvalue of  $T$ .*

## Remark

*For finite dimensional Hilbert space, the spectrum of  $T$  consist only the eigenvalues of  $T$ . However, in infinite dimensional Hilbert space, we may have no eigenvalues. For instance, consider the shift operator. Here, it has no eigenvalues, since if we consider  $(T - \lambda I)x = 0$ , we see that*

$$(-\lambda x_1, x_1 - \lambda x_2, x_2 - \lambda x_3, \dots) = (0, 0, 0, \dots) \implies x = (0, 0, 0, \dots)$$

*So,  $T$  does not have an eigenvalue. However, we showed that  $T$  is not invertible. So,  $T - 0I$  is not invertible and  $0 \in \sigma(T)$ . So, the spectrum of an operator does not only consist of eigenvalues.*



### Proof.

Suppose that  $\lambda$  is not an eigenvalue of  $T$ , then there is nonzero  $u \in H$  such that  $Tu = \lambda u$ . So, there is no nonzero  $u \in H$  such that  $(T - \lambda I)u = 0$ . So,  $\ker(T - \lambda I) = \{0\}$ . By rank-nullity theorem, we have

$$\begin{aligned}\dim H &= \dim \ker(T - \lambda I) + \dim \operatorname{Im}(T - \lambda I) \\ &= 0 + \dim \operatorname{Im}(T - \lambda I) \\ &= \dim \operatorname{Im}(T - \lambda I)\end{aligned}$$

So,  $\operatorname{Im}(T - \lambda I) = H$ . This means that  $(T - \lambda I)$  is surjective, and therefore, invertible. So,  $\lambda \in \rho(T)$ , and therefore,  $\lambda \notin \sigma(T)$ .  $\square$

## Proposition

Let  $\mathcal{H}$  be a Hilbert space. Let  $T : \mathcal{H} \rightarrow \mathcal{H}$  be linear operator. The following properties hold:

1.  $\sigma(I) = \{1\}$
2.  $\sigma(T^*) = \{\bar{\lambda} : \lambda \in \sigma(T)\}$
3. If  $T$  is invertible, then  $\sigma(T^{-1}) = \{\mu^{-1} : \mu \in \sigma(T)\}$
4. If  $T$  is self-adjoint, then  $\sigma(T) \subseteq \mathbb{R}$ .
5. If  $U : \mathcal{H} \rightarrow \mathcal{H}$  is invertible, then  $\sigma(UTU^{-1}) = \sigma(T)$

### Lemma

*If  $\mathcal{H}$  is a complex Hilbert space and  $U \in B(\mathcal{H})$  is unitary then*

$$\sigma(U) \subseteq \{\lambda \in \mathbb{C} : |\lambda| = 1\}$$

## Theorem

*Let  $\mathcal{H}$  be a complex Hilbert space and let  $T \in B(\mathcal{H})$ . If  $|\lambda| > \|T\|$  then  $\lambda \notin \sigma(T)$ .*

## Corollary

*If  $T \in B(H)$ , then  $\sigma(T)$  is bounded.*

### Proof.

If  $\lambda \in \sigma(T)$ , then, by the contrapositive of Theorem 4,  $|\lambda| \leq \|T\|$ .  
So,  $\lambda \in \overline{B(0, \|T\|)}$ . Hence,  $\sigma(T) \subseteq \overline{B(0, \|T\|)}$ , i.e.  $\sigma(T)$  is bounded. □

## Theorem

*If  $T \in B(H)$ , then  $\sigma(T) \subseteq \mathbb{C}$  is a closed set.*

## Corollary

*If  $T \in B(H)$ , then  $\sigma(T)$  is compact in  $\mathbb{C}$*



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We would like to find the spectrum of the unilateral shift,  $T : \ell^2 \rightarrow \ell^2$ , defined by

$$T(x_1, x_2, \dots) = (0, x_1, x_2, \dots)$$

Since  $T$  has been shown to have no eigenvalue, we will consider the eigenvalues of  $T^*$  instead and use the properties of the spectrum to have a lower bound for the spectrum.

Suppose that  $|\lambda| < 1$ , then consider the sequence

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We have

$$\sum_{n=0}^{\infty} |x_n|^2 = \sum_{n=0}^{\infty} (|\lambda|^n)^2 = \sum_{n=0}^{\infty} (|\lambda|^2)^n = \frac{1}{1 - |\lambda|^2} < \infty$$

Now, observe that

$$\begin{aligned} T_X &= T(1, \lambda, \lambda^2, \dots) \\ &= (\lambda, \lambda^2, \lambda^3, \dots) = \lambda(1, \lambda, \lambda^2, \dots) = \lambda x \end{aligned}$$

Now, observe that

$$\begin{aligned}Tx &= T(1, \lambda, \lambda^2, \dots) \\ &= (\lambda, \lambda^2, \lambda^3, \dots) = \lambda(1, \lambda, \lambda^2, \dots) = \lambda x\end{aligned}$$

So, every  $\lambda \in \mathbb{C}$  such that  $|\lambda| < 1$  is an eigenvalue of  $T^*$ .



Now, observe that

$$\begin{aligned}Tx &= T(1, \lambda, \lambda^2, \dots) \\ &= (\lambda, \lambda^2, \lambda^3, \dots) = \lambda(1, \lambda, \lambda^2, \dots) = \lambda x\end{aligned}$$

So, every  $\lambda \in \mathbb{C}$  such that  $|\lambda| < 1$  is an eigenvalue of  $T^*$ . Now, we have

$$\{\lambda : |\lambda| < 1\} = \{\bar{\lambda} : |\lambda| < 1\} \subseteq \sigma(T)$$

So,

$$\{\lambda \in \mathbb{C} : |\lambda| < 1\} \subseteq \sigma(T) \subseteq \{\lambda \in \mathbb{C} : |\lambda| \leq 1\}$$

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or

$$B(0, 1) \subseteq \sigma(T) \subseteq \overline{B(0, 1)}$$

So,

$$\{\lambda \in \mathbb{C} : |\lambda| < 1\} \subseteq \sigma(T) \subseteq \{\lambda \in \mathbb{C} : |\lambda| \leq 1\}$$

or

$$B(0, 1) \subseteq \sigma(T) \subseteq \overline{B(0, 1)}$$

Since  $\sigma(T)$  is closed, we have

$$\sigma(T) = \overline{B(0, 1)} = \{\lambda \in \mathbb{C} : |\lambda| \leq 1\}$$